

Modeling Cell-Cell Interactions in T-Cell Dependent Cytotoxicity (TDCC) Based Cancer Therapeutics

Takeda Oncology

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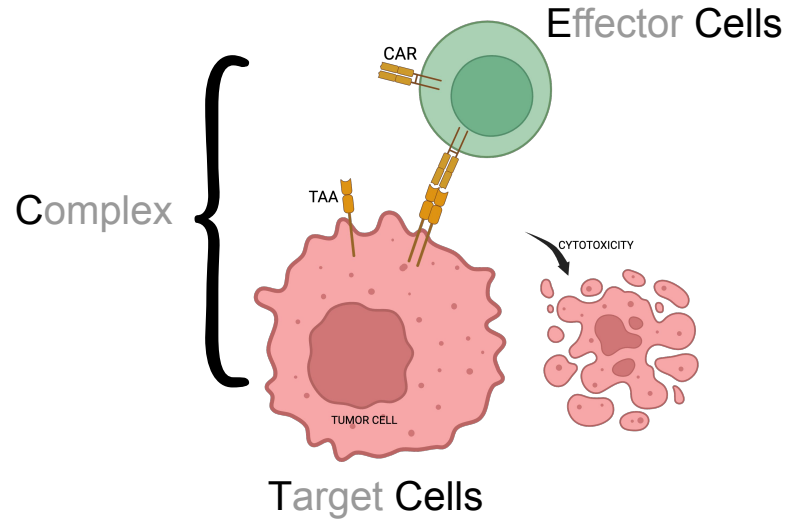
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Background

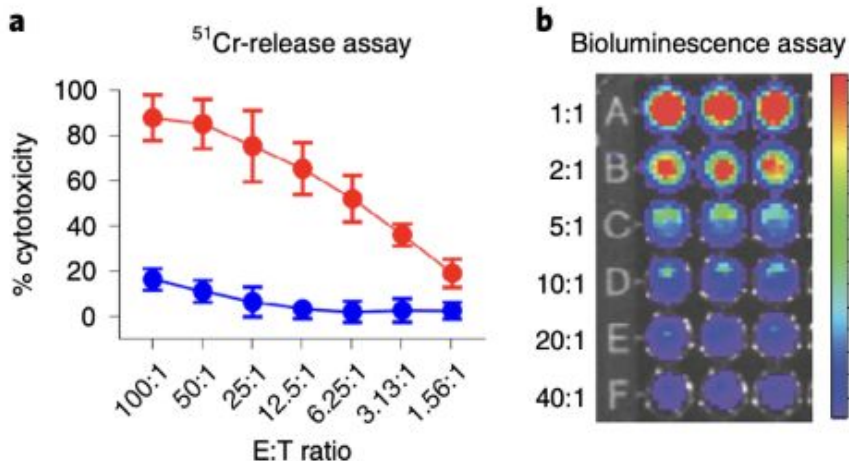
CAR T Cells (Chimeric Antigen Receptors) TAA (Tumor-Associated Antigen)



Measuring Target Cell Killing in CAR

- **Why Measuring E:T Ratios?**

- High E:T ratios → maximize signal-to-noise in vitro
- In vitro conditions ≠ physiological conditions
- Modeling needed to extrapolate to real E:T scenarios



Goals

- Can the **ratio between Effector and Target cells (E:T ratio)** reliably explain cytotoxicity effectiveness?
- What are **some other quantities** that explain the effectiveness?



Overview

Start with Agent-Based Modeling

Reflect cell-level behavior

Bridge to Ordinary Differential Equation System

Fit to ABM dynamics

Study the goal efficiently

Fit to Data

Fit ODEs to ABM data

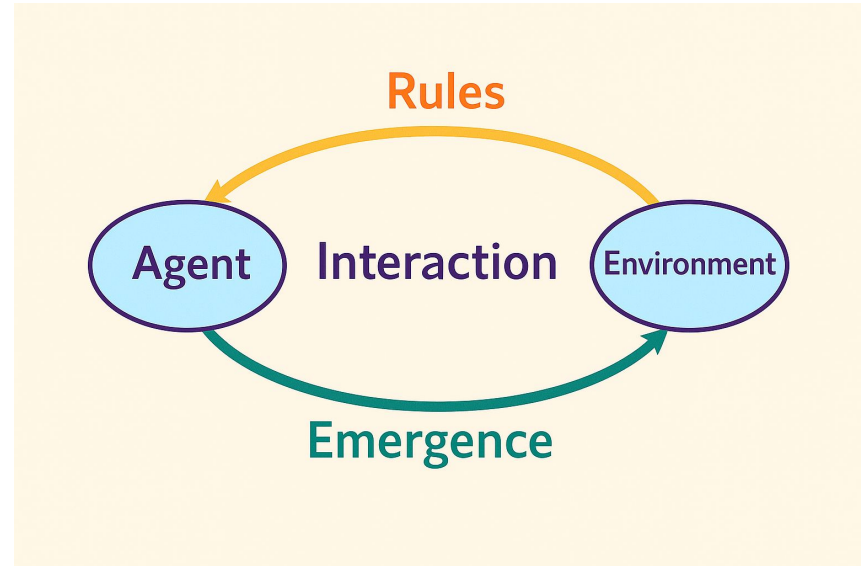
Fit to real data



Agent-Based Modeling Overview

“Simulate the system by simulating the parts.”

- Agent
- Interaction
- Environment



Agent-Based Model (ABM) Setup

Agent:

Effective Cell
Targeted Cell
E-T **Complex**

Interaction:

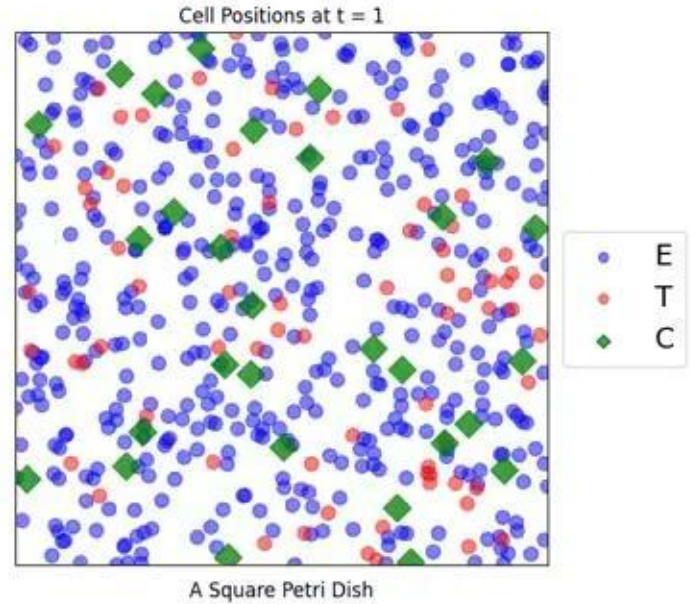
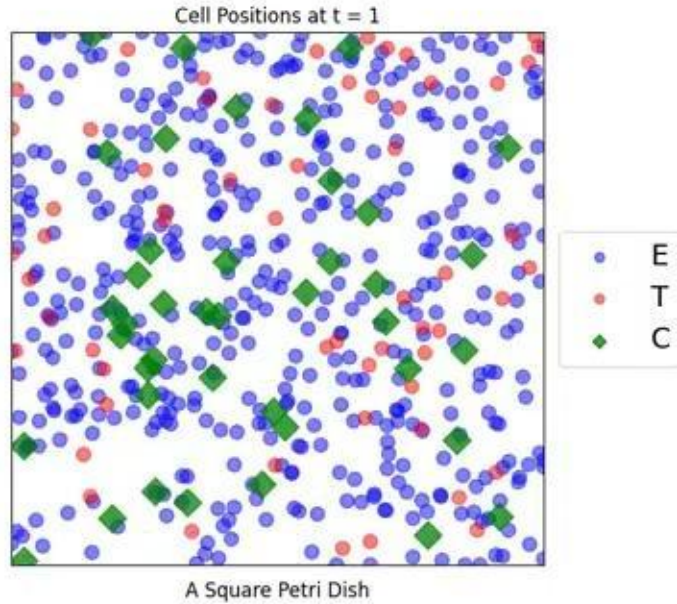
T: **Grow**
E: **Move, Bind**
C: **Unbind, Kill, Persist**

Environment:

Petri dish –
a '**Well**' with
boundary



ABM Simulations



- Simulate population behavior with simple rules
- Different outcomes emerge due to randomness

From ABM to ODE

- Run ABM simulations
- Record data
- Use ODE to fit the data

But Why ODE?

- Computation Efficiency
- Analytical tractability
- Reproducibility
- Ease of communication and integration



Ordinary Differential Equation System Overview

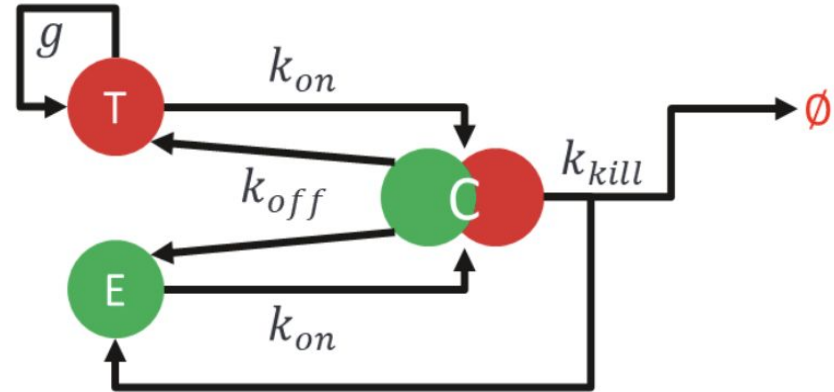
“Capture overall behavior with equations that track how things change.”

- (Almost) Same setup as ABM
- ... But with math equations!
- Solve the equations (versus: run the simulation)

$$\frac{dT}{dt} = gT - k_{on}E \cdot T + k_{off}C$$

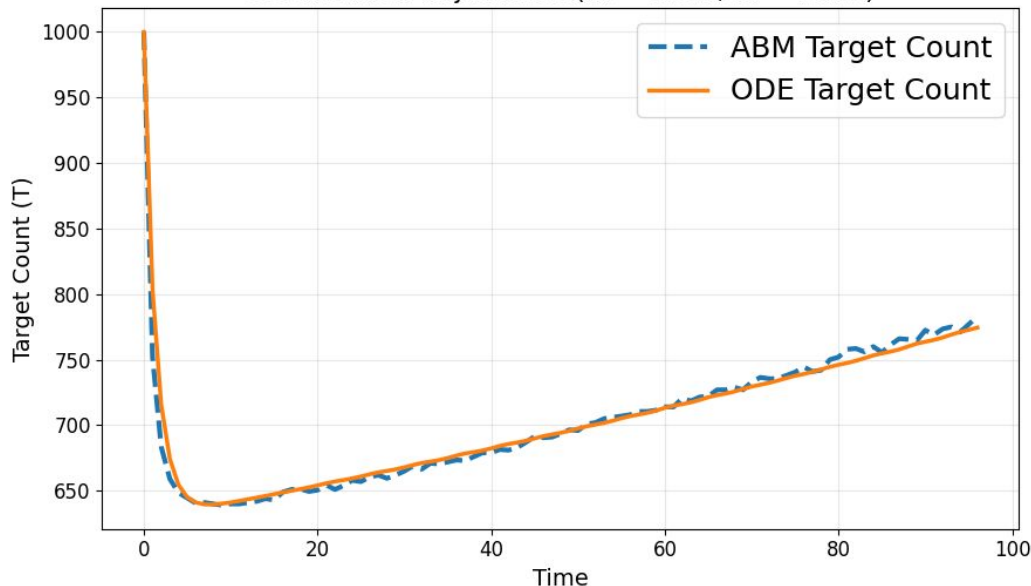
$$\frac{dE}{dt} = -k_{on}E \cdot T + k_{off}C + k_{kill}C$$

$$\frac{dC}{dt} = k_{on}E \cdot T - k_{off}C - k_{kill}C$$



Compare ABM and ODE Results

ODE vs ABM Trajectories ($E_0 = 1000$, $T_0 = 1000$)



- Strong match
- Slightly worse with global fit

We are now ready to study and answer the main questions.
Is E:T ratio important? What else are important?

Quantify Effectiveness: F-Metric

The F-metric compares how many Target cells are left **with** treatment (Effector cells) versus **without** them

$$F(t) = \frac{T(t, E_0)}{T(t, 0)} = \frac{\text{Target Left with Treatment}}{\text{Target Left without Treatment}}$$

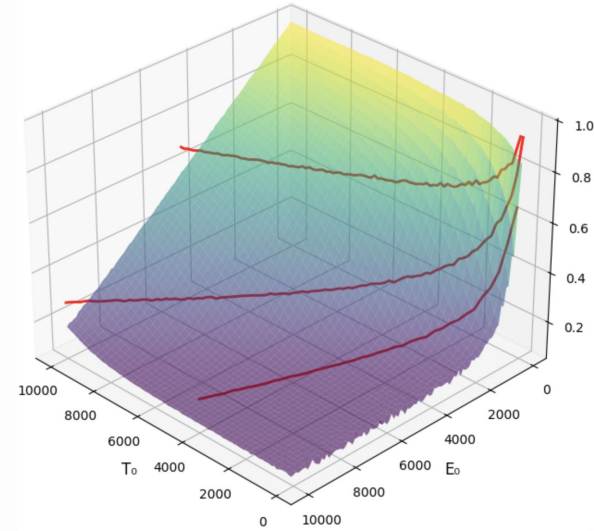
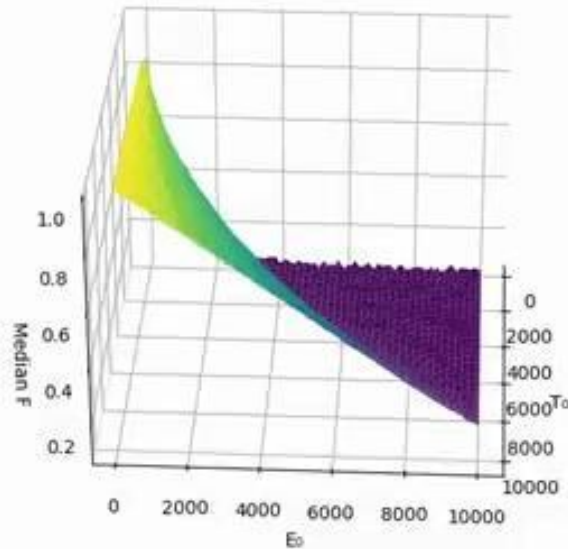
Higher F (close to 1), less successful...

Lower F (close to 0), more successful!



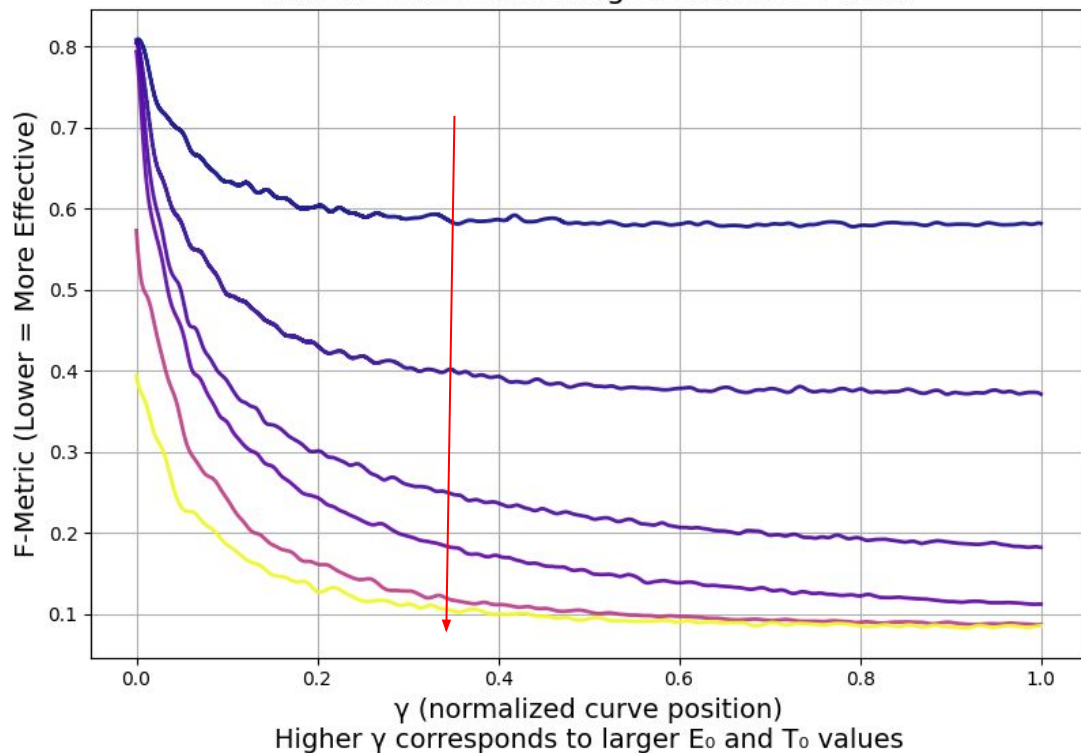
F-Metric Surface

“How does cytotoxicity vary across different initial Effector and Target counts?”

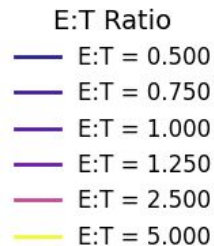


Slice Along Constant E:T Ratio with ABM

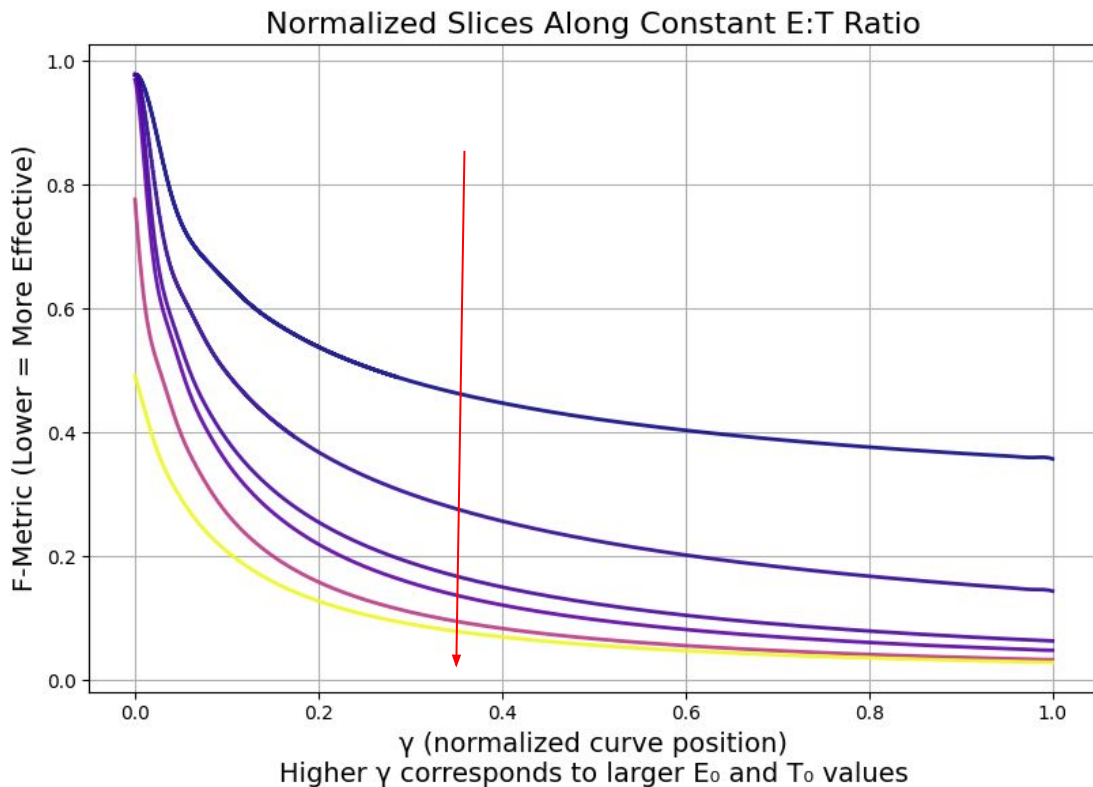
Normalized Slices Along Constant E:T Ratio



- More effective with higher E:T
- Curve is not flat



Slice Along Constant E:T Ratio With ODE



- Same shape as ABM
- More effective with higher E:T
- Curve is not flat

E:T Ratio

- E:T = 0.50
- E:T = 0.75
- E:T = 1.00
- E:T = 1.25
- E:T = 2.50
- E:T = 5.00



Answers to Main Questions

- Can the **ratio between Effector and Target cells (E:T ratio)** reliably explain cytotoxicity effectiveness?

Yes!

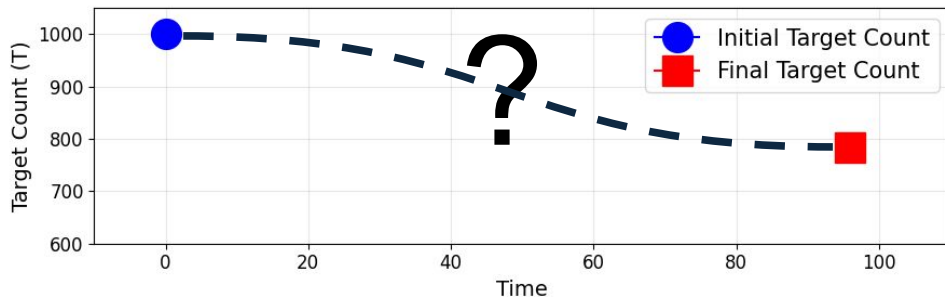
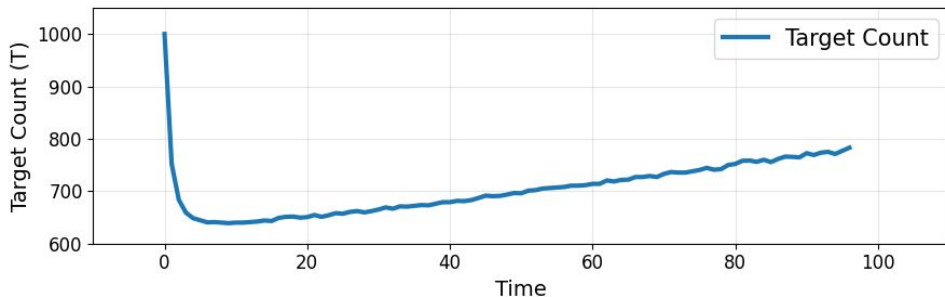
- What are **some other quantities** that explain the effectiveness?

Absolute count of effector and target cells

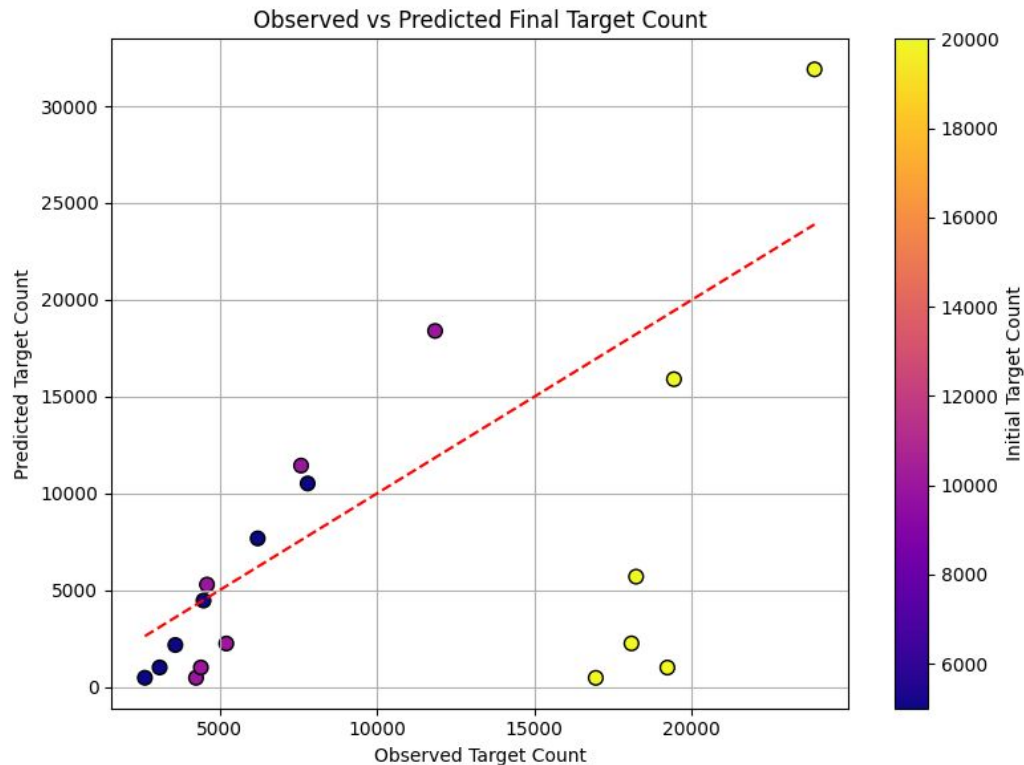


Fit to Lab Data

- 18 data points, representing 18 groups of trials:
 - 3 different initial target cell counts
 - 6 different E:T ratios
 - % killing information
- **No individual-cell or time-resolved data**



Fitting Result



- Point near diagonal means good fit
- Overall fit quality is not good
- **Model tends to overestimate importance of effector count**
- Need a model that can better align with lab data
- Are there anything else we can do with the data?

Hill Equation, An Empirical Alternative Approach

“How response increases with dose.”

- Model percentage killing (i.e. $1 - F$)
- **Captures saturation: response plateaus at high input**
- Includes a steepness factor (Hill coefficient) to model sensitivity
- Common in biology for modeling drug response or cell killing

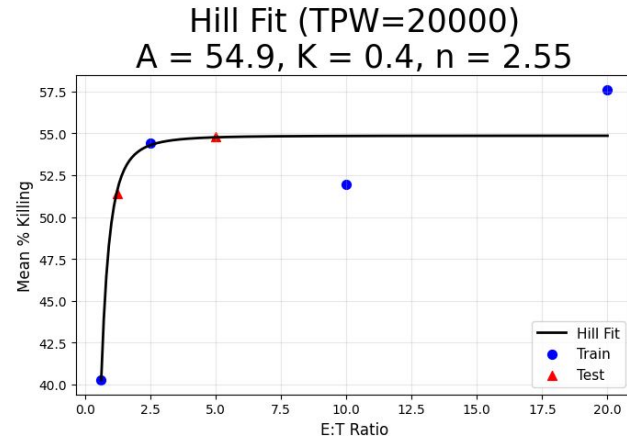
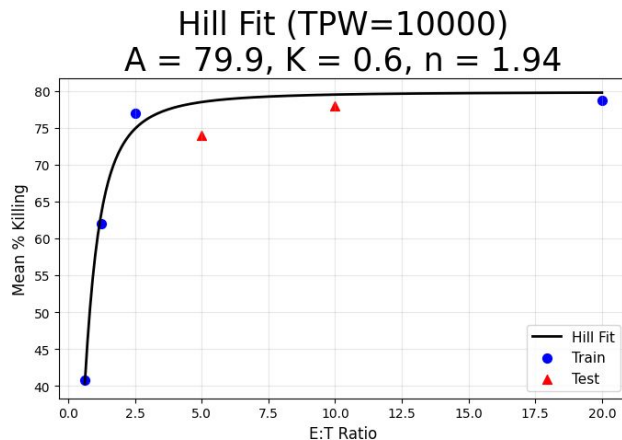
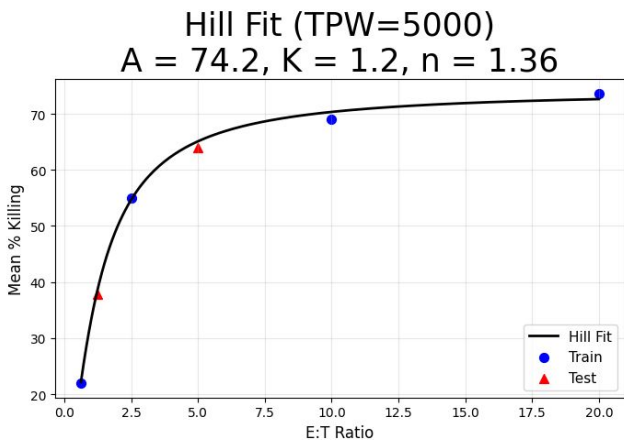


Data Fitting Setup

- 18 data points total, each point has:
 - Initial target cell count
 - E:T ratio
 - % Killing
- Data grouped by initial target cell count (3 groups), within each group :
 - 4 points for training ($\frac{2}{3}$)
 - 2 points for testing ($\frac{1}{3}$)
- Fit Hill equation on training data
- Evaluated fit performance on testing/overall data



Hill Equation



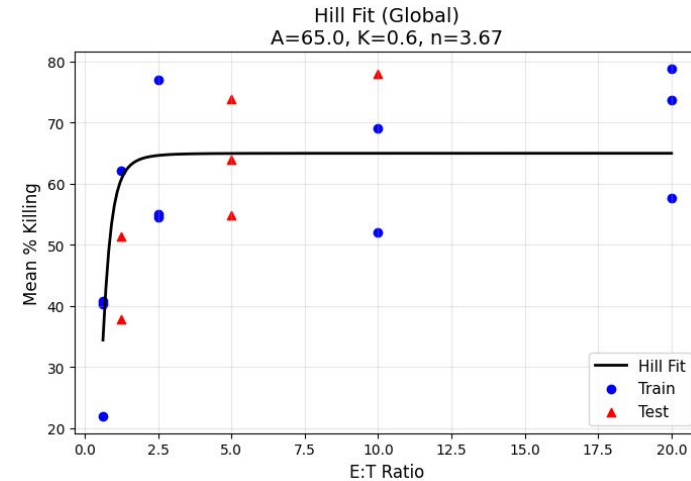
Observation:

- A: decrease with TPW (initial target cell count)
- K: decrease
- n: increase



Hill Equation Results

- Great fit even with tiny dataset with good predicting power
 - E:T ratio can predict treatment effectiveness, BUT,
 - Also need initial target cell count information



- A (max killing rate) ↓ with TPW: high tumor load overwhelms effector cells
- K (half-max point) ↓ with TPW: system becomes more sensitive to E:T ratio
- n (steepness) ↑ with TPW: crowded environment triggers more all-or-none killing behavior

Future work

- Alignment of ABM and ODE models with lab observations
- Use of Master Equation for probability distribution of cell counts
- Challenge of limited data; need for data-efficient modeling strategies



Thank you !

